5. Let $\mathcal{T}_1$ and $\mathcal{T}_2$ be two topologies on X such that both $(X, \mathcal{T}_1)$ and $(X, \mathcal{T}_2)$ are compact Hausdorff spaces. Prove that either $\mathcal{T}_1 = \mathcal{T}_2$ or they are not comparable.

Proof. Suppose $\mathcal{T}_1$ and $\mathcal{T}_2$ are comparable. Without loss of generality, let

$$\mathcal{T}_1 \subseteq \mathcal{T}_2.$$

Consider the identity map

$$I : (X, \mathcal{T}_2) \to (X, \mathcal{T}_1), \qquad I(x) := x.$$

Clearly, I is *bijective*.
For all $U \in \mathcal{T}_1$, we have

$$I^{-1}(U) = U \in \mathcal{T}_2.$$

Hence, I is *continuous*.
Since I is a bijective continuous map from a compact space to a Hausdorff space, I is a *homeomorphism*. So, I^{-1} is also continuous.
Thus, for all $U \in \mathcal{T}_2$, we have

$$(I^{-1})^{-1}(U) = I(U) = U \in \mathcal{T}_1.$$

Hence,

$$\mathcal{T}_2 \subseteq \mathcal{T}_1.$$

Therefore,

$$\boxed{\mathcal{T}_1 = \mathcal{T}_2.}$$

Hence, either $\mathcal{T}_1 = \mathcal{T}_2$ or they are not comparable. □

6. Prove that every compact subspace of a metric space is closed and bounded in that metric topology. Find an example of a metric space in which a closed and bounded subspace is not compact.

Proof. Let (X, d) be a metric space and let $K \subseteq X$ be compact.

We know that every metric space is Hausdorff and compact subspace of a Hausdorff space is closed. Since X is a metric Space it is Hausdorff. And K is a compact subspace of X, thus it is *closed* in X.

Let $a \in X$. Consider the collection

$$\mathcal{U} := \{B(a, n) : n \in \mathbb{N}\},$$

where

$$B(a, n) := \{x \in X : d(x, a) < n\}.$$

For all $x \in K, d(x, a)$ is finite; so, $\exists n \in \mathbb{N}$ such that $d(x, a) < n$; i.e., $x \in B(a, n)$. Thus, $\mathcal{U}$ is an open cover of A.

Since K is compact, there exists a finite subcover $\{B(a, n_1), \ldots, B(a, n_m)\}$. Therefore,

$$K \subseteq \bigcup_{i=1}^{m} B(a, n_i).$$

Let, $N := \max\{n_1, n_2, \ldots, n_m\}$. Since $B(a, n_i) \subseteq B(a, N)$ for every i, we get

$$K \subseteq \bigcup_{i=1}^{m} B(a, n_i) \subseteq B(a, N).$$

Therefore, K is *bounded*.

Hence,

$$\boxed{K \text{ is closed and bounded.}}$$

□

Example:

Consider the metric space $X := (0, 1)$ with the usual metric ($\because d(x, y) := |x - y|$), and let $A := X$.
Clearly, A is closed ($\because$ it is the entire space) and bounded ($\because \forall x \in A, 0 < x < 1$) in X
However, consider the collection

$$\mathcal{U} = \left\{ \left(\frac{1}{n}, 1 \right) : n \in \mathbb{N}, n \geq 2 \right\}.$$

Each member of $\mathcal{U}$ is open in the subspace $(0, 1)$. And,

$$(0, 1) = \bigcup_{n=2}^{\infty} \left(\frac{1}{n}, 1 \right).$$

Thus, $\mathcal{U}$ is an open cover of A.
For every finite subcollection $\left\{ \left(\frac{1}{n_1}, 1 \right), \left(\frac{1}{n_2}, 1 \right), \ldots, \left(\frac{1}{n_k}, 1 \right) \right\}$. Let $N = \max\{n_1, n_2, \ldots, n_k\}$.
Since $n_i \leq N$ for every $i = 1, \ldots, k$, we have

$$\left(\frac{1}{n_i}, 1 \right) \subseteq \left(\frac{1}{N}, 1 \right).$$

Therefore,

$$\bigcup_{i=1}^{k} \left(\frac{1}{n_i}, 1 \right) \subseteq \left(\frac{1}{N}, 1 \right) \subsetneq (0, 1).$$

It cannot cover $(0, 1)$, since points in $(0, 1/N]$ remain uncovered.
Hence, $\mathcal{U}$ has no finite subcover.
Therefore,

$$\boxed{(0, 1) \text{ is closed and bounded but not compact in } (0, 1).}$$